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Getting started

Forecasting has fascinated people for thousands of years, sometimes being considered a sign of divine inspiration, and sometimes being seen as a criminal activity. The Jewish prophet Isaiah wrote in about 700 BC

Tell us what the future holds, so we may know that you are gods.
(Isaiah 41:23)

One hundred years later, in ancient Babylon, forecasters would foretell the future based on the distribution of maggots in a rotten sheep's liver. By 300 BC, people wanting forecasts would journey to Delphi in Greece to consult the Oracle, who would provide her predictions while intoxicated by ethylene vapours. Forecasters had a tougher time under the emperor Constantine, who issued a decree in AD357 forbidding anyone "to consult a soothsayer, a mathematician, or a forecaster ... May curiosity to foretell the future be silenced forever." A similar ban on forecasting occurred in England in 1736 when it became an offence to defraud by charging money for predictions. The punishment was three months' imprisonment with hard labour!

The varying fortunes of forecasters arise because good forecasts can seem almost magical, while bad forecasts may be dangerous. Consider the following famous predictions about computing.

- *I think there is a world market for maybe five computers.*
(Chairman of IBM, 1943)
- *Computers in the future may weigh no more than 1.5 tons.*
(Popular Mechanics, 1949)

- *There is no reason anyone would want a computer in their home.*
(President, DEC, 1977)

The last of these was made only three years before IBM produced the first personal computer. Not surprisingly, you can no longer buy a DEC computer. Forecasting is obviously a difficult activity, and businesses that do it well have a big advantage over those whose forecasts fail.

In this book, we will explore the most reliable methods for producing forecasts. The emphasis will be on methods that are replicable and testable, and have been shown to work.

1/1 What can be forecast?

Forecasting is required in many situations: deciding whether to build another power generation plant in the next five years requires forecasts of future demand; scheduling staff in a call centre next week requires forecasts of call volumes; stocking an inventory requires forecasts of stock requirements. Forecasts can be required several years in advance (for the case of capital investments), or only a few minutes beforehand (for telecommunication routing). Whatever the circumstances or time horizons involved, forecasting is an important aid to effective and efficient planning.

Some things are easier to forecast than others. The time of the sunrise tomorrow morning can be forecast very precisely. On the other hand, tomorrow's lotto numbers cannot be forecast with any accuracy. The predictability of an event or a quantity depends on several factors including:

1. how well we understand the factors that contribute to it;
2. how much data are available;
3. whether the forecasts can affect the thing we are trying to forecast.

For example, forecasts of electricity demand can be highly accurate because all three conditions are usually satisfied. We have a good idea on the contributing factors: electricity demand is driven largely by temperatures, with smaller effects for calendar variation such as holidays, and economic conditions. Provided there is a sufficient history of data on electricity demand and weather conditions, and we have the skills to

develop a good model linking electricity demand and the key driver variables, the forecasts can be remarkably accurate.

On the other hand, when forecasting currency exchange rates, only one of the conditions is satisfied: there is plenty of available data. However, we have a very limited understanding of the factors that affect exchange rates, and forecasts of the exchange rate have a direct effect on the rates themselves. If there are well-publicised forecasts that the exchange rate will increase, then people will immediately adjust the price they are willing to pay and so the forecasts are self-fulfilling. In a sense the exchange rates become their own forecasts. This is an example of the “efficient market hypothesis”. Consequently, forecasting whether the exchange rate will rise or fall tomorrow is about as predictable as forecasting whether a tossed coin will come down as a head or a tail. In both situations, you will be correct about 50% of the time whatever you forecast. In situations like this, forecasters need to be aware of their own limitations, and not claim more than is possible.

Often in forecasting, a key step is knowing when something can be forecast accurately, and when forecasts will be no better than tossing a coin. Good forecasts capture the genuine patterns and relationships which exist in the historical data, but do not replicate past events that will not occur again. In this book, we will learn how to tell the difference between a random fluctuation in the past data that should be ignored, and a genuine pattern that should be modelled and extrapolated.

Many people wrongly assume that forecasts are not possible in a changing environment. Every environment is changing, and a good forecasting model captures the way in which things are changing. Forecasts rarely assume that the environment is unchanging. What is normally assumed is that *the way in which the environment is changing* will continue into the future. That is, a highly volatile environment will continue to be highly volatile; a business with fluctuating sales will continue to have fluctuating sales; and an economy that has gone through booms and busts will continue to go through booms and busts. A forecasting model is intended to capture the way things move, not just where things are. As Abraham Lincoln said, “If we could first know where we are and whither we are tending, we could better judge what to do and how to do it”.

Forecasting situations vary widely in their time horizons, factors determining actual outcomes, types of data patterns, and many other aspects. Forecasting methods can be very simple such as using the most recent observation as a forecast (which is called the “naïve method”), or highly complex such as neural nets and econometric systems of simultaneous equations. Sometimes, there will be no data available at all. For example, we may wish to forecast the sales of a new product in its first year, but there are obviously no data to work with. In situations like this, we use judgmental forecasting, discussed in Chapter 3. The choice of method depends on what data are available and the predictability of the quantity to be forecast.

1/2 Forecasting, planning and goals

Forecasting is a common statistical task in business, where it helps to inform decisions about the scheduling of production, transportation and personnel, and provides a guide to long-term strategic planning. However, business forecasting is often done poorly, and is frequently confused with planning and goals. They are three different things.

Forecasting is about predicting the future as accurately as possible, given all of the information available, including historical data and knowledge of any future events that might impact the forecasts.

Goals are what you would like to have happen. Goals should be linked to forecasts and plans, but this does not always occur. Too often, goals are set without any plan for how to achieve them, and no forecasts for whether they are realistic.

Planning is a response to forecasts and goals. Planning involves determining the appropriate actions that are required to make your forecasts match your goals.

Forecasting should be an integral part of the decision-making activities of management, as it can play an important role in many areas of a company. Modern organizations require short-term, medium-term and long-term forecasts, depending on the specific application.

Short-term forecasts are needed for the scheduling of personnel, production and transportation. As part of the scheduling process, forecasts of demand are often also required.

Medium-term forecasts are needed to determine future resource requirements, in order to purchase raw materials, hire personnel, or buy machinery and equipment.

Long-term forecasts are used in strategic planning. Such decisions must take account of market opportunities, environmental factors and internal resources.

An organization needs to develop a forecasting system that involves several approaches to predicting uncertain events. Such forecasting systems require the development of expertise in identifying forecasting problems, applying a range of forecasting methods, selecting appropriate methods for each problem, and evaluating and refining forecasting methods over time. It is also important to have strong organizational support for the use of formal forecasting methods if they are to be used successfully.

1/3 Determining what to forecast

In the early stages of a forecasting project, decisions need to be made about what should be forecast.

For example, if forecasts are required for items in a manufacturing environment, it is necessary to ask whether forecasts are needed for:

1. every product line, or for groups of products?
2. every sales outlet, or for outlets grouped by region, or only for total sales?
3. weekly data, monthly data or annual data?

It is also necessary to consider the forecasting horizon. Will forecasts be required for one month in advance, for 6 months, or for ten years? Different types of models will be necessary depending on what forecast horizon is most important.

How frequently are forecasts required? Forecasts that need to be produced frequently are better done using an automated system than with methods that require careful manual work.

It is worth spending time talking to the people who will use the forecasts to ensure that you understand their needs, and

how the forecasts are to be used, before embarking on extensive work in producing the forecasts.

Once it has been determined what forecasts are required, it is then necessary to find or collect the data on which the forecasts will be based. The data required for forecasting may already exist. These days, a lot of data are recorded, and the forecaster's task is often to identify where and how the required data are stored. The data may include sales records of a company, the historical demand for a product, or the unemployment rate for a geographical region. A large part of a forecaster's time can be spent in locating and collating the available data prior to developing suitable forecasting methods.

1/4 Forecasting data and methods

The appropriate forecasting methods depend largely on what data are available.

If there are no data available, or if the data available are not relevant to the forecasts, then *qualitative forecasting* methods must be used. These methods are not purely guesswork—there are well-developed structured approaches to obtaining good forecasts without using historical data. These methods are discussed in Chapter 3.

Quantitative forecasting can be applied when two conditions are satisfied:

1. numerical information about the past is available;
2. it is reasonable to assume that some aspects of the past patterns will continue into the future.

There is a wide range of quantitative forecasting methods, often developed within specific disciplines for specific purposes. Each method has its own properties, accuracies, and costs that must be considered when choosing a specific method.

Most quantitative forecasting problems use either time series data (collected at regular intervals over time) or cross-sectional data (collected at a single point in time).

Cross-sectional forecasting

With cross-sectional data, we are wanting to predict the value of something we have not observed, using the information on the

cases that we have observed. Examples of cross-sectional data include:

- House prices for all houses sold in 2011 in a particular area. We are interested in predicting the price of a house not in our data set using various house characteristics: position, number of bedrooms, age, etc.
- Fuel economy data for a range of 2009 model cars. We are interested in predicting the carbon footprint of a vehicle not in our data set using information such as the size of the engine and the fuel efficiency of the car.

Example 1.1 Car emissions

Table 1.1 gives some data on 2009 model cars, each of which has an automatic transmission and an engine size under 2 liters.

Model	Litres (litres)	City (mpg)	Highway (mpg)	Carbon (tons CO ₂)
Chevrolet Aveo	1.6	25	34	6.6
Chevrolet Aveo 5	1.6	25	34	6.6
Honda Civic	1.8	25	36	6.3
Honda Civic Hybrid	1.3	40	45	4.4
Honda Fit	1.5	27	33	6.1
Honda Fit	1.5	28	35	5.9
Hyundai Accent	1.6	26	35	6.3
Kia Rio	1.6	26	35	6.1
Nissan Versa	1.8	27	33	6.3
Nissan Versa	1.8	24	32	6.8
Pontiac G3 Wave	1.6	25	34	6.6
Pontiac G3 Wave 5	1.6	25	34	6.6
Pontiac Vibe	1.8	26	31	6.6
Saturn Astra 2DR Hatchback	1.8	24	30	6.8
Saturn Astra 4DR Hatchback	1.8	24	30	6.8
Scion xD	1.8	26	32	6.6
Toyota Corolla	1.8	27	35	6.1
Toyota Matrix	1.8	25	31	6.6
Toyota Prius	1.5	48	45	4.0
Toyota Yaris	1.5	29	35	5.9

Table 1.1: Fuel economy and carbon footprints for 2009 model cars with automatic transmissions and small engines. City and Highway represent fuel economy while driving in the city and on the highway.

A forecaster may wish to predict the carbon footprint (tons of CO₂ per year) for other similar vehicles that are not included in the above table. It is necessary to first estimate the effects of the predictors (number of cylinders, size of engine, and fuel economy) on the variable to be forecast (carbon footprint). Then, provided that we know the predictors for a car not in the table, we can forecast its carbon footprint.

Cross-sectional models are used when the variable to be forecast exhibits a relationship with one or more other predictor variables. The purpose of the cross-sectional model is to describe the form of the relationship and use it to forecast values of the forecast variable that have not been observed. Under this model, any change in predictors will affect the output of the system in a predictable way, assuming that the relationship does not change. Models in this class include regression models, additive models, and some kinds of neural networks. These models are discussed in Chapters 4, 5 and 9.

Some people use the term “predict” for cross-sectional data and “forecast” for time series data (see below). In this book, we will not make this distinction—we will use the words interchangeably.

Time series forecasting

Time series data are useful when you are forecasting something that is changing over time (e.g., stock prices, sales figures, profits, etc.). Examples of time series data include:

- Daily IBM stock prices
- Monthly rainfall
- Quarterly sales results for Amazon
- Annual Google profits

Anything that is observed sequentially over time is a time series. In this book, we will only consider time series that are observed at regular intervals of time (e.g., hourly, daily, weekly, monthly, quarterly, annually). Irregularly spaced time series can also occur, but are beyond the scope of this book.

When forecasting time series data, the aim is to estimate how the sequence of observations will continue into the future. Figure 1.1 shows the quarterly Australian beer production from 1992 to the third quarter of 2008.

The blue lines show forecasts for the next two years. Notice how the forecasts have captured the seasonal pattern seen in the historical data and replicated it for the next two years. The dark shaded region shows 80% prediction intervals. That is, each future value is expected to lie in the dark shaded region with a probability of 80%. The light shaded region shows 95% prediction intervals. These prediction intervals are a very useful way of displaying the uncertainty in forecasts. In this case, the

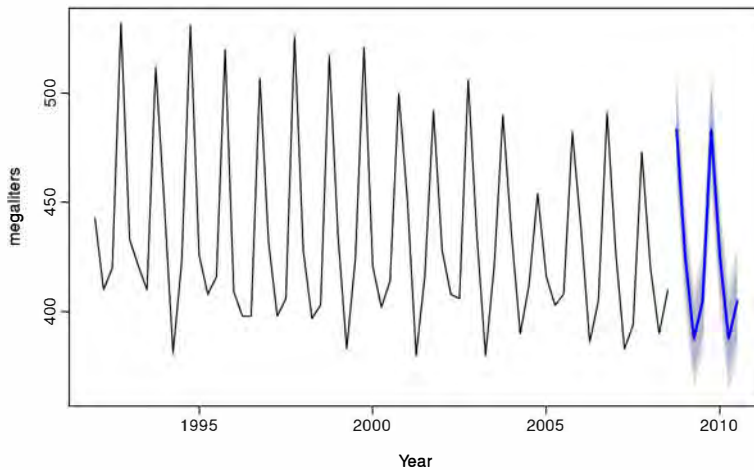


Figure 1.1: Australian quarterly beer production: 1992Q1–2008Q3, with two years of forecasts.

forecasts are expected to be very accurate, hence the prediction intervals are quite narrow.

Time series forecasting uses only information on the variable to be forecast, and makes no attempt to discover the factors which affect its behavior. Therefore it will extrapolate trend and seasonal patterns, but it ignores all other information such as marketing initiatives, competitor activity, changes in economic conditions, and so on.

Time series models used for forecasting include ARIMA models, exponential smoothing and structural models. These models are discussed in Chapters 6, 7 and 8.

Predictor variables and time series forecasting

Predictor variables can also be used in time series forecasting. For example, suppose we wish to forecast the hourly electricity demand (ED) of a hot region during the summer period. A model with predictor variables might be of the form

$$\text{ED} = f(\text{current temperature, strength of economy, population, time of day, day of week, error}).$$

The relationship is not exact—there will always be changes in electricity demand that cannot be accounted for by the predictor variables. The “error” term on the right allows for random variation and the effects of relevant variables that not included in the model. We call this an “explanatory model”

because it helps explain what causes the variation in electricity demand.

Because the electricity demand data form a time series, we could also use a time series model for forecasting. In this case, a suitable time series forecasting equation is of the form

$$ED_{t+1} = f(ED_t, ED_{t-1}, ED_{t-2}, ED_{t-3}, \dots, \text{error}),$$

where t is the present hour, $t + 1$ is the next hour, $t - 1$ is the previous hour, $t - 2$ is two hours ago, and so on. Here, prediction of the future is based on past values of a variable, but not on external variables which may affect the system. Again, the “error” term on the right allows for random variation and the effects of relevant variables that are not included in the model.

There is also a third type of model which combines the features of the above two models. For example, it might be given by

$$ED_{t+1} = f(ED_t, \text{current temperature, time of day, day of week, error}).$$

These types of mixed models have been given various names in different disciplines. They are known as dynamic regression models, panel data models, longitudinal models, transfer function models, and linear system models (assuming f is linear). These models are discussed in Chapter 9.

An explanatory model is very useful because it incorporates information about other variables, rather than only historical values of the variable to be forecast. However, there are several reasons a forecaster might select a time series model rather than an explanatory model. First, the system may not be understood, and even if it was understood it may be extremely difficult to measure the relationships that are assumed to govern its behavior. Second, it is necessary to know or forecast the future values of the various predictors in order to be able to forecast the variable of interest, and this may be too difficult. Third, the main concern may be only to predict what will happen, not to know why it happens. Finally, the time series model may give more accurate forecasts than an explanatory or mixed model.

The model to be used in forecasting depends on the resources and data available, the accuracy of the competing models, and how the forecasting model is to be used.

Notation

For cross-sectional data, we will use the subscript i to indicate a specific observation. For example, y_i will denote the i th observation in a data set. We will also use N to denote the total number of observations in the data set.

For time series data, we will use the subscript t instead of i . For example, y_t will denote the observation at time t . We will use T to denote the number of observations in a time series.

When we are making general comments that could be applicable to either cross-sectional or time series data, we will tend to use i and N .

1/5 Some case studies

The following four cases are from our consulting practice and demonstrate different types of forecasting situations and the associated problems that often arise.

Case 1 The client was a large company manufacturing disposable tableware such as napkins and paper plates. They needed forecasts of each of hundreds of items every month. The time series data showed a range of patterns, some with trends, some seasonal, and some with neither. At the time, they were using their own software, written in-house, but it often produced forecasts that did not seem sensible. The methods that were being used were the following:

1. average of the last 12 months data;
2. average of the last 6 months data;
3. prediction from a straight line regression over the last 12 months;
4. prediction from a straight line regression over the last 6 months;
5. prediction obtained by a straight line through the last observation with slope equal to the average slope of the lines connecting last year's and this year's values;
6. prediction obtained by a straight line through the last observation with slope equal to the average slope of the lines connecting last year's and this year's values, where the average is taken only over the last 6 months.

They required us to tell them what was going wrong and to modify the software to provide more accurate forecasts. The software was written in COBOL making it difficult to do any sophisticated numerical computation.

Case 2 In this case, the client was the Australian federal government who needed to forecast the annual budget for the Pharmaceutical Benefit Scheme (PBS). The PBS provides a subsidy for many pharmaceutical products sold in Australia, and the expenditure depends on what people purchase during the year. The total expenditure was around A\$7 billion in 2009 and had been underestimated by nearly \$1 billion in each of the two years before we were asked to assist with developing a more accurate forecasting approach.

In order to forecast the total expenditure, it is necessary to forecast the sales volumes of hundreds of groups of pharmaceutical products using monthly data. Almost all of the groups have trends and seasonal patterns. The sales volumes for many groups have sudden jumps up or down due to changes in what drugs are subsidised. The expenditures for many groups also have sudden changes due to cheaper competitor drugs becoming available.

Thus we needed to find a forecasting method that allowed for trend and seasonality if they were present, and at the same time was robust to sudden changes in the underlying patterns. It also needed to be able to be applied automatically to a large number of time series.

Case 3 A large car fleet company asked us to help them forecast vehicle re-sale values. They purchase new vehicles, lease them out for three years, and then sell them. Better forecasts of vehicle sales values would mean better control of profits; understanding what affects resale values may allow leasing and sales policies to be developed in order to maximize profits.

At the time, the resale values were being forecast by a group of specialists. Unfortunately, they saw any statistical model as a threat to their jobs and were uncooperative in providing information. Nevertheless, the company provided a large amount of data on previous vehicles and their eventual resale values.

Case 4 In this project, we needed to develop a model for forecasting weekly air passenger traffic on major domestic routes for one of Australia's leading airlines. The company required forecasts of passenger numbers for each major domestic route and for each class of passenger (economy class, business class and first class). The company provided weekly traffic data from the previous six years.

Air passenger numbers are affected by school holidays, major sporting events, advertising campaigns, competition behaviour, etc. School holidays often do not coincide in different Australian cities, and sporting events sometimes move from one city to another. During the period of the historical data, there was a major pilots' strike during which there was no traffic for several months. A new cut-price airline also launched and folded. Towards the end of the historical data, the airline had trialled a redistribution of some economy class seats to business class, and some business class seats to first class. After several months, however, the seat classifications reverted to the original distribution.

1/6 *The basic steps in a forecasting task*

A forecasting task usually involves five basic steps.

Step 1: Problem definition. Often this is the most difficult part of forecasting. Defining the problem carefully requires an understanding of the way the forecasts will be used, who requires the forecasts, and how the forecasting function fits within the organization requiring the forecasts. A forecaster needs to spend time talking to everyone who will be involved in collecting data, maintaining databases, and using the forecasts for future planning.

Step 2: Gathering information. There are always at least two kinds of information required: (a) statistical data, and (b) the accumulated expertise of the people who collect the data and use the forecasts. Often, it will be difficult to obtain enough historical data to be able to fit a good statistical model. However, occasionally, very old data will be less useful due to changes in the system being forecast.

Step 3: Preliminary (exploratory) analysis. Always start by graphing the data. Are there consistent patterns? Is there a significant trend? Is seasonality important? Is there evidence of the presence of business cycles? Are there any outliers in the data that need to be explained by those with expert knowledge? How strong are the relationships among the variables available for analysis? Various tools have been developed to help with this analysis. These are discussed in Chapters 2 and 6.

Step 4: Choosing and fitting models. The best model to use depends on the availability of historical data, the strength of relationships between the forecast variable and any explanatory variables, and the way the forecasts are to be used. It is common to compare two or three potential models. Each model is itself an artificial construct that is based on a set of assumptions (explicit and implicit) and usually involves one or more parameters which must be “fitted” using the known historical data. We will discuss regression models (Chapters 4 and 5), exponential smoothing methods (Chapter 7), Box-Jenkins ARIMA models (Chapter 8), and a variety of other topics including dynamic regression models, neural networks and vector autoregression in Chapter 9.

Step 5: Using and evaluating a forecasting model. Once a model has been selected and its parameters estimated, the model is used to make forecasts. The performance of the model can only be properly evaluated after the data for the forecast period have become available. A number of methods have been developed to help in assessing the accuracy of forecasts. There are also organizational issues in using and acting on the forecasts. A brief discussion of some of these issues is in Chapter 2.

1/7 *The statistical forecasting perspective*

The thing we are trying to forecast is unknown (or we wouldn't be forecasting it), and so we can think of it as a *random variable*. For example, the total sales for next month could take a range of possible values, and until we add up the actual sales at the end of the month we don't know what the value will be. So, until we know the sales for next month, it is a random quantity.

Because next month is relatively close, we usually have a good idea what the likely sales values could be. On the other hand, if we are forecasting the sales for the same month next year, the possible values it could take are much more variable. In most forecasting situations, the variation associated with the thing we are forecasting will shrink as the event approaches. In other words, the further ahead we forecast, the more uncertain we are.

When we obtain a forecast, we are estimating the *middle* of the range of possible values the random variable could take. Very often, a forecast is accompanied by a prediction interval giving a *range* of values the random variable could take with relatively high probability. For example, a 95% prediction interval contains a range of values which should include the actual future value with probability 95%.

A forecast is always based on some observations. Suppose we denote all the information we have observed as $\mathcal{I}$ and we want to forecast y_i . We then write $y_i|\mathcal{I}$ meaning “the random variable y_i given what we know in $\mathcal{I}$ ”. The set of values that this random variable could take, along with their relative probabilities, is known as the “probability distribution” of $y_i|\mathcal{I}$. In forecasting, we call this the “forecast distribution”.

When we talk about the “forecast”, we usually mean the average value of the forecast distribution, and we put a “hat” over y to show this. Thus, we write the forecast of y_i as $\hat{y}_i$, meaning the average of the possible values that y_i could take given everything we know. Occasionally, we will use $\hat{y}_i$ to refer to the *median* (or middle value) of the forecast distribution instead.

With time series forecasting, it is often useful to specify exactly what information we have used in calculating the forecast. Then we will write, for example, $\hat{y}_{i|t-1}$ to mean the forecast of y_t taking account of all previous observations $(y_1, \dots, y_{t-1})$. Similarly, $\hat{y}_{T+h|T}$ means the forecast of y_{T+h} taking account of $y_1, \dots, y_T$ (i.e., an h -step forecast taking account of all observations up to time T).

1/8 Exercises

- 1.1 For each of the four case studies in Section 1/5, what sort of data is involved: time series or cross-sectional data?
- 1.2 For cases 3 and 4 in Section 1/5, list the possible predictor variables that might be useful, assuming that the relevant data are available.
- 1.3 For case 3 in Section 1/5, describe the five steps of forecasting in the context of this project.

1/9 Further reading

- Armstrong, J. S., ed. (2001). *Principles of forecasting: a handbook for researchers and practitioners*. Boston, MA: Kluwer Academic Publishers.
- Ord, J. K. and R. Fildes (2012). *Principles of business forecasting*. South-Western College Pub.